

Wimukthi Madushan

B.Sc. Engineering Honours
Department of Computer Science and Engineering,
University of Moratuwa

 [GitHub Profile](#)

 [LinkedIn Profile](#)

 wimukthi.21@cse.mrt.ac.lk

 [Website](#)

 [\(+94\) 740631472](tel:+94740631472)

SUMMARY

A Final Year Computer Science and Engineering student with a strong interest in AI-based software development, machine learning, and LLM integration. Skilled in problem-solving and passionate about building intelligent, user-focused solutions that combine modern software engineering with AI technologies. Eager to learn, adapt, and contribute to innovative real-world projects through hands-on experience.

Areas of Interest: AI-based Software Development, LLM Integration, Machine Learning, Deep Learning, Front-end and Back-end Development, Mobile Development, Database Management

EXPERIENCE

Software Engineering Intern
Scienter Technologies (Pvt) Ltd

Jan 2025 – Mar 2025

- Contributed to the development of web applications using Angular, .Net frameworks and SQL.
- Participated in daily standups, code reviews, and agile sprint planning.
- Gained hands-on experience in full-stack development and software deployment practices.

EDUCATION

University of Moratuwa

Bachelor of Science (Hons) in Engineering, Computer Science and Engineering

Aug. 2022 - Present

CGPA: 3.51/4

Govt. Science College Matale

GCE Advanced Level - Physical Science Stream - 3A passes with a Z-score of 2.2875

Jan. 2015 - Aug. 2020

PROJECTS

- **Final Year Research Project: Dengue Case Research (Ongoing)** 2025
Developing a Spatial-Temporal Graph Neural Network (ST-GNN) model to predict dengue cases across Sri Lankan districts, analyzing spatial connections and temporal trends between regions to identify outbreak patterns and inter-district relationships.
- **JobAlign.AI: CV Matching and RAG System (Ongoing)** 2025
Building a React Native mobile application that calculates the similarity score between a candidate's CV and job description. Integrating a RAG-based chatbot for companies to query CVs, along with an email sending system for candidate communication. The system uses Clerk authentication, FastAPI backend with Hugging Face models, and MongoDB for data storage.
Technologies: React Native, Clerk, FastAPI, MongoDB, Hugging Face, RAG, SMTP, Expo, Docker
Organization **GitHub:** [GitHub](#) 

- MCP Server For my CV and Send Email (Retrieval-Augmented Generation) 2025**
 Implemented a Model Context Protocol (MCP) server capable of answering resume-related queries (RAG Server) and sending email notifications. Initially designed with a separate REST API layer, later simplified into a single MCP server with HTTP endpoints exposed via Swagger/OpenAPI. Configured IDE integration with MCP to allow direct connectivity and provided a simple frontend for interaction.
Technologies: Python, FastMCP, FastAPI, OpenAI API, SMTP, Swagger/OpenAPI, Next.js, Docker
Frontend: [GitHub](#)  **Backend:** [GitHub](#) 
- Sanchaara (Backend) 2025**
 Built a mobile application using Flutter and designed a scalable backend with Java Spring Boot. Implemented microservice architecture with asynchronous communication via Kafka, and containerized all services using Docker and Docker Compose for consistent deployment.
Role: Backend Developer – Responsible for designing and developing the entire backend system.
Technologies: Java, Spring Boot, Flutter, Kafka, Docker, MongoDB, PostgreSQL
Frontend: [GitHub](#)  **Backend:** [GitHub](#) 
- Potato Disease Classification Using CNN (Deep Learning) 2025**
 Developed a deep learning model using Convolutional Neural Networks (CNNs) to classify potato plant diseases from leaf images. The project involved data preprocessing, model training, evaluation, and deployment. Achieved high accuracy in detecting common potato diseases such as early blight and late blight.
Technologies: Python, Flask, FastAPI, Tensorflow, CNN, Next JS, NumPy, Matplotlib
Frontend: [GitHub](#)  **Backend:** [GitHub](#) 
- Deep Learning Image Classifier (Deep Learning)  2025**
 Developed an image classification system to distinguish between cats and dogs using two different approaches. The first model was trained from scratch using a custom CNN architecture, while the second leveraged transfer learning with a pre-trained model (Xception). Achieved high accuracy through data augmentation, optimization techniques, and model evaluation strategies.
Technologies: Python, Flask, Tensorflow, keras, Pandas, Deep Learning, openCV
- Laptop Price Predictor (ML)  2025**
 A machine learning web application that predicts laptop prices based on key hardware specifications such as CPU, GPU, RAM, screen type, and more. The system includes data preprocessing, feature engineering, and model training for accurate predictions. Built with a user-friendly Flask interface, allowing users to input specs and instantly view estimated prices.
Technologies: Python, Flask, Scikit-learn, Pandas, Machine Learning
- Microservice Deployment on AWS with CI/CD (DevOps)  2025**
 Designed and developed a simple microservice application using Java Spring Boot. The application was containerized with Docker and deployed on an AWS EC2 instance. Integrated with a MySQL database hosted on AWS RDS, and automated the CI/CD pipeline using GitHub Actions for seamless deployment and updates.
Technologies: Java, Spring Boot, Docker, AWS EC2, AWS RDS, MySQL, GitHub Actions
Related Project: [Node.js Application Deploy on AWS with CI/CD](#)
[Node.js Application Deploy EC2 Docker and CI/CD](#)
- Interactive Library Management System (Web and mobile)  2024**
 Fifth semester Software Engineering project, I contributed to the development of an interactive library management system. This project involves creating web applications for efficient book management and user interaction.
 Contribution: Frontend, Backend, Database, Documentation

Technologies: React React Native, Node.js, Express, MySQL, Expo

- **Airline Reservation System**

2024

third semester Database Management System project, I contributed to the development of an Airline Reservation System. This project involves creating web applications for efficient Airline management and user interaction.

Contribution: Frontend, Backend, Database, Documentation

Technologies: React, Flask, MySQL

CORE SKILLS

- **Web Development:** HTML, CSS, React, tailwind CSS, Node.js, Express, Next js, Angular, Spring Boot, FAST API, .Net , UI Libraries(Radix UI, Ant Design, Chakra UI)
- **Mobile Development:** React Native
- **Database Management:** MySQL, MongoDB, PostgreSQL
- **Version Control:** Git, GitHub
- **Programming Languages:** C, C++, Python, Java, JavaScript, TypeScript, Python
- **Deployment:** AWS, Render, Vercel
- **Tools:** VSCode, IntelliJ IDEA, Docker, Xilinx Vivado

SOFT SKILLS

- Effective communication and teamwork skills gained through group projects and collaborations.
- Adaptability and quick learning ability in adopting new technologies and methodologies.
- Time management and organizational skills developed through balancing academic and project commitments.

LICENSES & CERTIFICATIONS

- **Introduction to Front-End Development (Meta)**  *Nov 2023*
- **Introduction to Back-End Development (Meta)**  *Nov 2023*
- **Introduction to Databases for Back-End Development (Meta)**  *Dec 2023*
- **Version Control (Meta)**  *Dec 2023*
- **Programming with JavaScript (Meta)**  *Jan 2024*
- **Advanced React (Meta)**  *Jan 2024*

REFERENCES

Prof. (Ms) Meedeniya D.A.

Senior Lecturer

Department of Computer Science & Engineering
ing

University of Moratuwa

Sri Lanka

dulanim@cse.mrt.ac.lk

Ms. D M S M M Dissanayaka

MSc, BSc

Senior Software Engineer

Postsig Inc

San Francisco, CA

sachini@postsig.com

Dr. Kutila Gunasekera

Senior Lecturer

Department of Computer Science & Engineering

University of Moratuwa

Sri Lanka

Mobile: +94 77 1423877

kutilla@cse.mrt.ac.lk